Zestaw 2

Zadanie 1. Rozwiązywanie równań, `Solve`, `Plot`, `Epilog`

Rozwiąż równanie: $x^3 + x^2 - 4x - 4 = 0$

In[]:=
```
f[x_] := x^3 + x^2 - 4 x - 4
```

In[]:=
```
? Solve
```

Out[]=

Symbol

Solve[*expr*, *vars*] attempts to solve the system *expr* of equations or inequalities for the variables *vars*.

Solve[*expr*, *vars*, *dom*] solves over the domain

dom. Common choices of *dom* are Reals, Integers, and Complexes.

In[]:=
```
solutions = Solve[f[x] == 0, x]
```

Out[]=
```
{{x → -2}, {x → -1}, {x → 2}}
```

Wyrysuj funkcję $f(x) = x^3 + x^2 - 4x - 4$ używając polecania `Plot` w zakresie $-4 \le x \le 4$

In[]:=
```
? Plot
```

Out[]=

Symbol

Plot[f, {x, x_{min}, x_{max}}] generates a plot of f as a function of x from x_{min} to x_{max}.

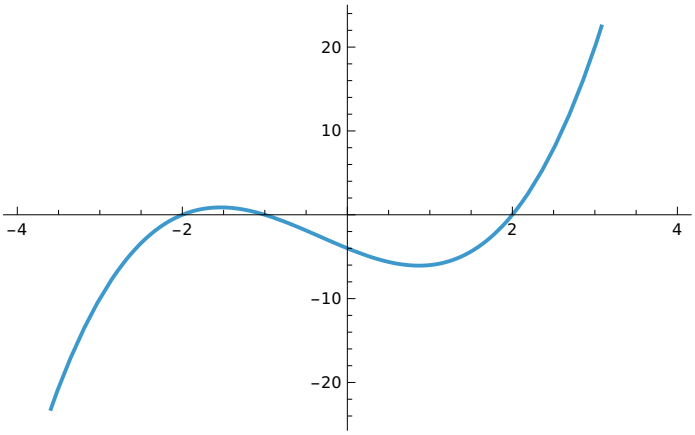
Plot[{f_1, f_2, ...}, {x, x_{min}, x_{max}}] plots several functions f_i.

Plot[{..., w[f_i], ...}, ...] plots f_i with features defined by the symbolic wrapper w.

Plot[..., {x} ∈ *reg*] takes the variable x to be in the geometric region *reg*.

In[•]:=
```
Plot[f[x], {x, -4, 4}]
```

Out[•]=



Na wykresie zaznacz rozwiązania równania, w formie czerwonych punktów. Podpisz osie wykresu

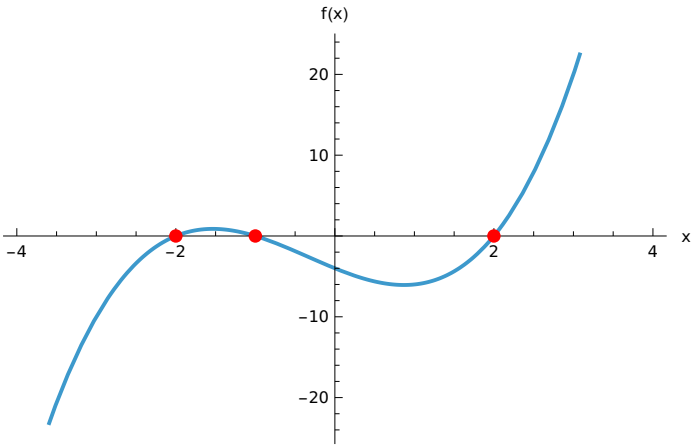
In[•]:=
```
GetSolutionPoints[f_, solutions_] := Map[
  x ↦ {x, f[x]},
  Values[Flatten[solutions]]
 ]
```

In[•]:=
```
solutionPoints = GetSolutionPoints[f, solutions]
```

Out[•]= {{-2, 0}, {-1, 0}, {2, 0}}

In[•]:=
```
Plot[
 f[x], {x, -4, 4},
 Epilog → {Red, PointSize[Large], Point[solutionPoints]},
 AxesLabel → {"x", "f(x)"}
]
```

Out[•]=



Zadanie 2. Rozwiązywanie równań, liczba rozwiązań, Solve, Plot, Reduce, Epilog, Manipulate

Zdefiniuj funkcje $f(x) = x^3 + x^2 - 4x - 4$ (jak w poprzednim zadaniu) oraz $g(x) = x^2 + a$, gdzie a jest parametrem rzeczywistym.

```
In[•]:= f[x_] := x^3 + x^2 - 4 x - 4
```

```
In[•]:= g[a_, x_] := x^2 + a
```

Rozwiąż równanie: $f(x) = g(x)$

```
In[•]:= Solve[f[x] == g[a, x], x, Reals]
```

```
Out[•]= {{x → Root[-4 - a - 4 #1 + #1^3 &, 1] if condition},
  {x → Root[-4 - a - 4 #1 + #1^3 &, 2] if condition}, {x → Root[-4 - a - 4 #1 + #1^3 &, 3] if condition}}
```

1) Dla $a = 0$, podaj rozwiązania (1) w $\mathbb{R}$ (zbiór liczb rzeczywistych)

```
In[•]:= a = 0;
solutions = Solve[f[x] == g[a, x], x, Reals]
```

```
Out[•]= {{x → 2.38...}}
```

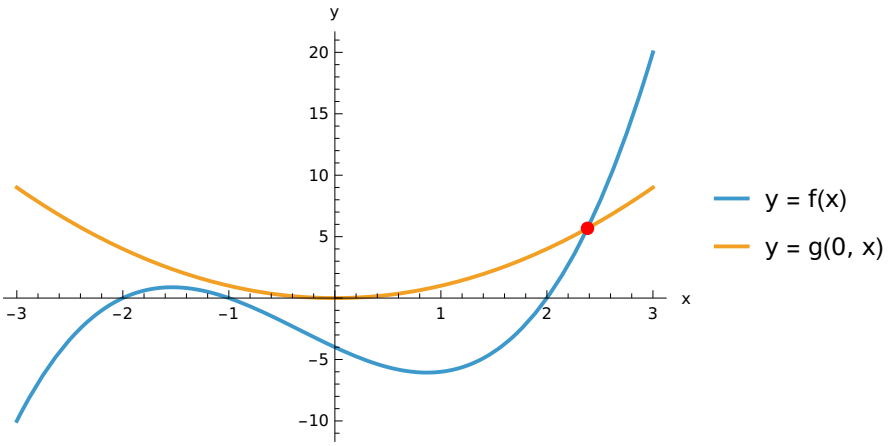
Przedstaw funkcje $f(x)$ oraz $g(x)$ na wykresie

```
In[•]:= PlotFunctions[a_, aStr_, solutions_] := Plot[
    {f[x], g[a, x]},
    {x, -3, 3},
    Epilog → {PointSize[Large], Red, Point[GetSolutionPoints[f, solutions]]},
    PlotLegends → {"y = f(x)", "y = g(" <> aStr <> ", x)"},
    AxesLabel → {"x", "y"}
   ];
```

```
In[•]:= PlotFunctions[a_, solutions_] := PlotFunctions[a, ToString[a], solutions]
```

In[•]:=
```
PlotFunctions[a, solutions]
```

Out[•]=



Ile jest punktów przecięcia obu funkcji?

In[•]:=
```
Length[Flatten[solutions]]
```

Out[•]= 1

2) Dla $a = -1$, podaj rozwiązania (1) w $\mathbb{R}$

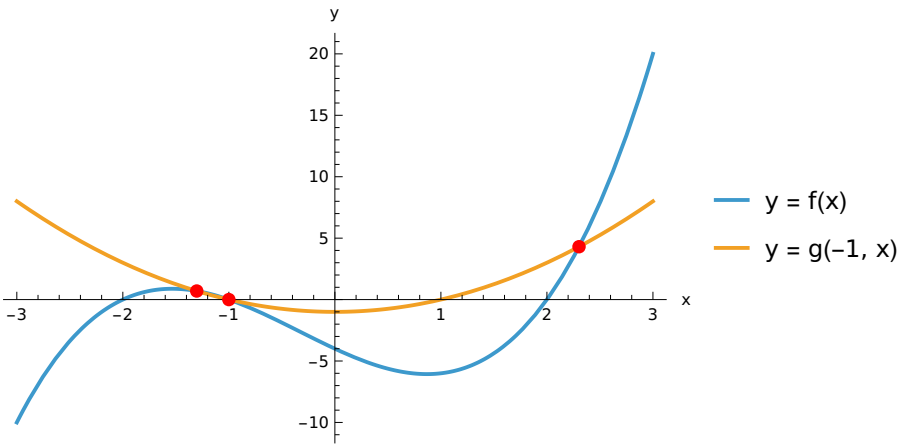
In[•]:=
```
a = -1;
solutions = Solve[f[x] == g[a, x], x, Reals]
```

Out[•]= $\left\{\{x \to -1\}, \left\{x \to \frac{1}{2}\left(1 - \sqrt{13}\right)\right\}, \left\{x \to \frac{1}{2}\left(1 + \sqrt{13}\right)\right\}\right\}$

Przedstaw graficznie rozwiązania równania

In[•]:=
```
PlotFunctions[a, solutions]
```

Out[•]=



3) Dla dowolnego a, określ przedziały a, dla których równanie (1) ma określoną liczbę rozwiązań

Przedstaw graficznie każdą sytuację, w której przy zmianie wartości a zmienia się liczba rozwiązań,

skorzystaj z Plot.
Rozwiązania (1) przedstaw na wykresie w formie czerwonych punktów.

Wskazówka: W punkcie 3) wydostań warunki na obecność danego rozwiązania używając funkcji `Part`. Te warunki to nierówności na a.
Używając funkcji `Reduce` należy je rozwiązać. Można użyć funkcji `FullSimplify` by uprościć wyrażenia.

```
In[•]:= Clear[a];
        solutionsForA = Solve[f[x] == g[a, x], x, Reals]
```

Out[•]= $\left\{\left\{x \to \text{Root}\left[-4 - a - 4\,\#1 + \#1^3\ \&,\ 1\right] \text{ if } \textit{condition}\right\},\right.$
$\left.\left\{x \to \text{Root}\left[-4 - a - 4\,\#1 + \#1^3\ \&,\ 2\right] \text{ if } \textit{condition}\right\}, \left\{x \to \text{Root}\left[-4 - a - 4\,\#1 + \#1^3\ \&,\ 3\right] \text{ if } \textit{condition}\right\}\right\}$

```
In[•]:= ExtractCondition[expression_] := FirstCase[
          expression,
          ConditionalExpression[_, c_] :> c,
          Missing[], {0, -1}
        ]
```

```
In[•]:= inequalities = Map[e ↦ ExtractCondition[e[[1]]], solutionsForA]
```

Out[•]= $\left\{\frac{4}{9}\left(-9 - 4\sqrt{3}\right) < a < \frac{4}{9}\left(-9 + 4\sqrt{3}\right) \,||\, a > \frac{4}{9}\left(-9 + 4\sqrt{3}\right) \,||\, a < \frac{4}{9}\left(-9 - 4\sqrt{3}\right),\right.$
$\left.\frac{4}{9}\left(-9 - 4\sqrt{3}\right) < a < \frac{4}{9}\left(-9 + 4\sqrt{3}\right),\ \frac{4}{9}\left(-9 - 4\sqrt{3}\right) < a < \frac{4}{9}\left(-9 + 4\sqrt{3}\right)\right\}$

```
In[•]:= ? Reduce
```

Out[•]=

Symbol

Reduce[*expr*, *vars*] reduces the statement *expr* by

solving equations or inequalities for *vars* and eliminating quantifiers.

Reduce[*expr*, *vars*, *dom*] does the reduction over the

domain *dom*. Common choices of *dom* are Reals, Integers, and Complexes.

3 rozwiązania

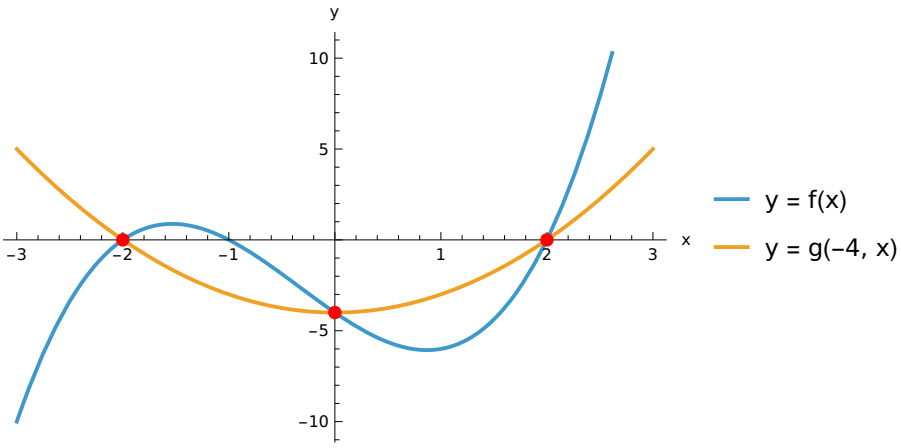
```
In[•]:= threeSolutions = FullSimplify[Reduce[inequalities, {a}]]
```

Out[•]= $-\frac{4}{9}\left(9 + 4\sqrt{3}\right) < a < \frac{4}{9}\left(-9 + 4\sqrt{3}\right)$

In[•]:=
```
SolveAndPlot[a_, aStr_] := Module[
   {solutions},
   solutions = Solve[f[x] == g[a, x], {x}, Reals];
   PlotFunctions[a, aStr, solutions]
  ];
SolveAndPlot[a_] := SolveAndPlot[a, ToString[a]]
```

In[•]:=
```
SolveAndPlot[-4]
```

Out[•]=



2 rozwiązania

Zauważmy, że:

In[•]:=
```
inequalities[[2]] === inequalities[[3]]
```

Out[•]=
```
True
```

In[•]:=
```
Reduce[{
   Not[inequalities[[1]]],
   inequalities[[2]]
  }]
```

Out[•]=
```
False
```

In[•]:=
```
noSolutions = FullSimplify[Reduce[{
     Not[inequalities[[1]]],
     Not[inequalities[[2]]]
    }]]
```

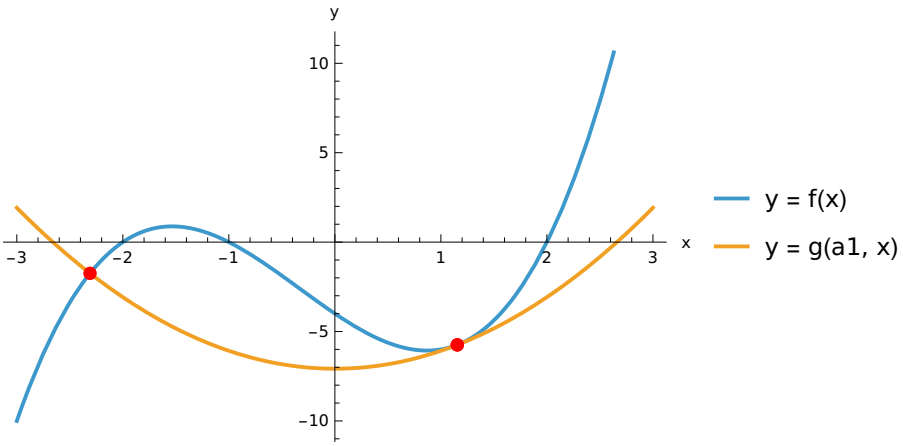
Out[•]=
$$36 + 16\sqrt{3} + 9\,a == 0 \;||\; 9\,(4 + a) == 16\sqrt{3}$$

In[•]:=
```
{a1, a2} = Values[Flatten[Solve[noSolutions]]]
```

Out[•]=
$$\left\{-\frac{4}{9}\left(9 + 4\sqrt{3}\right),\ \frac{4}{9}\left(-9 + 4\sqrt{3}\right)\right\}$$

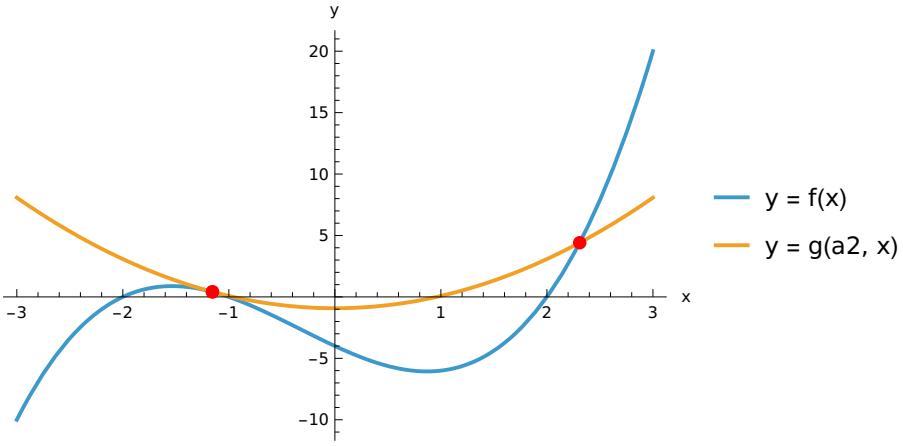
In[•]:= SolveAndPlot[a1, "a1"]

Out[•]=



In[•]:= SolveAndPlot[a2, "a2"]

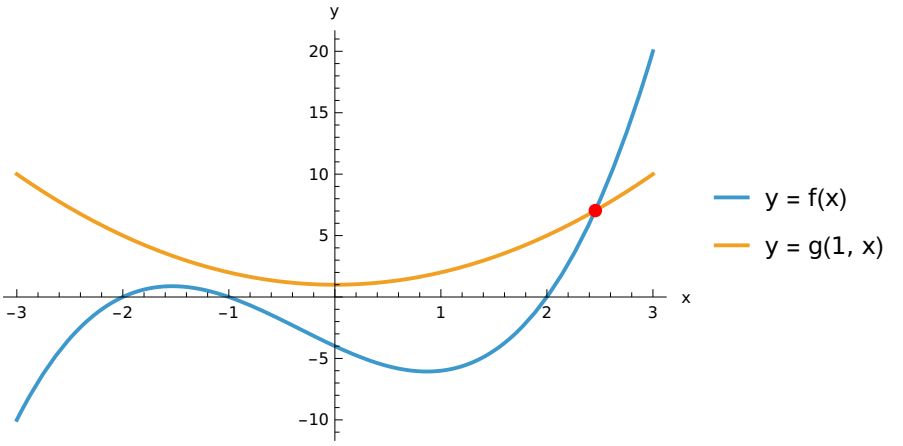
Out[•]=



1 rozwiązanie

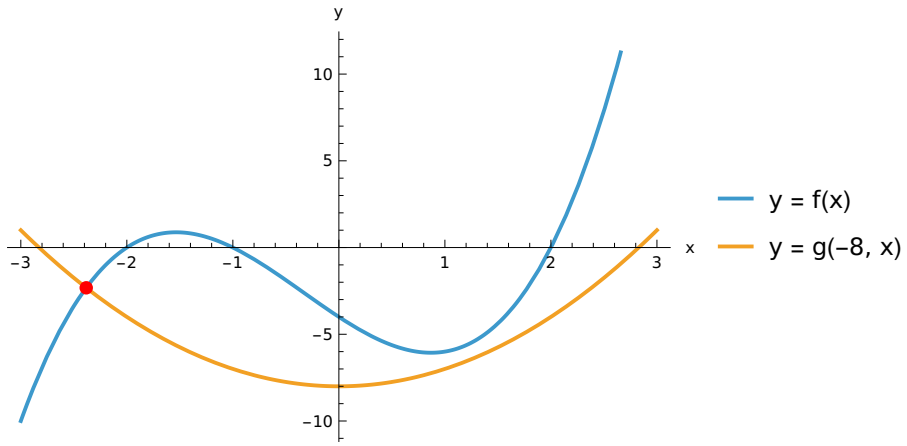
In[•]:= SolveAndPlot[1]

Out[•]=



In[•]:=
```
SolveAndPlot[-8]
```

Out[•]=



4) Skorzystaj z funkcji Manipulate, by przedstawić wykres z 3)

Zakres a zdefiniuj od -10 do 5.

In[•]:=
```
? Manipulate
```

Out[•]=

Symbol

Manipulate[*expr*, {u, u_{min}, u_{max}}] generates a version of

expr with controls added to allow interactive manipulation of the value of u.

Manipulate[*expr*, {u, u_{min}, u_{max}, du}] allows the value of u to vary between u_{min} and u_{max} in steps du.

Manipulate[*expr*, {{u, u_{init}}, u_{min}, u_{max}, ...}] takes the initial value of u to be u_{init}.

Manipulate[*expr*, {{u, u_{init}, u_{lbl}}, ...}] labels the controls for u with u_{lbl}.

Manipulate[*expr*, {u, {u_1, u_2, ...}}] allows u to take on discrete values u_1, u_2, ...

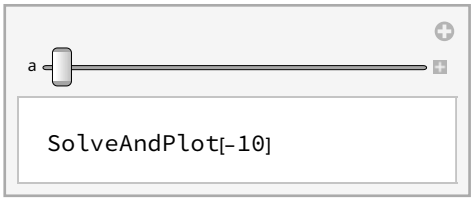
Manipulate[*expr*, {u, ...}, {v, ...}, ...] provides controls to manipulate each of the u, v, ...

Manipulate[*expr*, $c_u \to$ {u, ...}, $c_v \to$ {v, ...}, ...]

links the controls to the specified controllers on an external device.

In[•]:=
```
Manipulate[SolveAndPlot[a], {a, -10, 5}]
```

Out[•]=



Zadanie 3. Rozwiązywanie równań, dziedzina rozwiązań,

Solve

Sprawdź jakie są rozwiązania równania $a\,x^2 + b\,x + c = 0$ przy użyciu funkcji `Solve`:

```
In[•]:= p[x_] := a x^2 + b x + c
```

1) bez specyfikacji dziedziny

```
In[•]:= Solve[p[x] == 0, x]
```

$$\text{Out[•]=}\quad \left\{\left\{x \to \frac{-b - \sqrt{b^2 - 4\,a\,c}}{2\,a}\right\}, \left\{x \to \frac{-b + \sqrt{b^2 - 4\,a\,c}}{2\,a}\right\}\right\}$$

Jaka jest domyślna dziedzina dla równań w funkcji Solve?

Odp: **Liczby zespolone**

2) z dziedziną w zbiorze liczb rzeczywistych, $\mathbb{R}$

```
In[•]:= Solve[p[x] == 0, x, Reals]
```

$$\text{Out[•]=}\quad \left\{\left\{x \to -\frac{b}{2\,a} - \frac{1}{2}\sqrt{\frac{b^2 - 4\,a\,c}{a^2}} \text{ if } \textit{condition}\right\}, \left\{x \to -\frac{b}{2\,a} + \frac{1}{2}\sqrt{\frac{b^2 - 4\,a\,c}{a^2}} \text{ if } \textit{condition}\right\}\right\}$$

To samo sprawdź dla równia $a\,x^3 + b\,x^2 + c\,x + d$

```
In[•]:= p[x_] := a x^3 + b x^2 + c x + d
```

In[•]:=
```
Solve[p[x] == 0, x]
```

Out[•]=
$$\left\{\left\{x \to -\frac{b}{3\,a} - \frac{2^{1/3}\left(-b^2+3\,a\,c\right)}{3\,a\left(-2\,b^3+9\,a\,b\,c-27\,a^2\,d+\sqrt{4\left(-b^2+3\,a\,c\right)^3+\left(-2\,b^3+9\,a\,b\,c-27\,a^2\,d\right)^2}\right)^{1/3}} + \frac{\left(-2\,b^3+9\,a\,b\,c-27\,a^2\,d+\sqrt{4\left(-b^2+3\,a\,c\right)^3+\left(-2\,b^3+9\,a\,b\,c-27\,a^2\,d\right)^2}\right)^{1/3}}{3\times 2^{1/3}\,a}\right\},\right.$$

$$\left\{x \to -\frac{b}{3\,a} + \frac{\left(1+i\,\sqrt{3}\right)\left(-b^2+3\,a\,c\right)}{3\times 2^{2/3}\,a\left(-2\,b^3+9\,a\,b\,c-27\,a^2\,d+\sqrt{4\left(-b^2+3\,a\,c\right)^3+\left(-2\,b^3+9\,a\,b\,c-27\,a^2\,d\right)^2}\right)^{1/3}} - \frac{\left(1-i\,\sqrt{3}\right)\left(-2\,b^3+9\,a\,b\,c-27\,a^2\,d+\sqrt{4\left(-b^2+3\,a\,c\right)^3+\left(-2\,b^3+9\,a\,b\,c-27\,a^2\,d\right)^2}\right)^{1/3}}{6\times 2^{1/3}\,a}\right\},$$

$$\left.\left\{x \to -\frac{b}{3\,a} + \frac{\left(1-i\,\sqrt{3}\right)\left(-b^2+3\,a\,c\right)}{3\times 2^{2/3}\,a\left(-2\,b^3+9\,a\,b\,c-27\,a^2\,d+\sqrt{4\left(-b^2+3\,a\,c\right)^3+\left(-2\,b^3+9\,a\,b\,c-27\,a^2\,d\right)^2}\right)^{1/3}} - \frac{\left(1+i\,\sqrt{3}\right)\left(-2\,b^3+9\,a\,b\,c-27\,a^2\,d+\sqrt{4\left(-b^2+3\,a\,c\right)^3+\left(-2\,b^3+9\,a\,b\,c-27\,a^2\,d\right)^2}\right)^{1/3}}{6\times 2^{1/3}\,a}\right\}\right\}$$

In[•]:=
```
Solve[p[x] == 0, x, Reals]
```

Out[•]=
```
{{x → Root[d + c #1 + b #1^2 + a #1^3 &, 1] if condition},
 {x → Root[d + c #1 + b #1^2 + a #1^3 &, 2] if condition},
 {x → Root[d + c #1 + b #1^2 + a #1^3 &, 3] if condition}}
```

Rozważ równanie dla wielomianu n-tego stopnia

Rozważ równanie dla wielomianu n-tego stopnia postaci $x^n + a_{n-1}\,x^{n-1} + \ldots + a_1\,x + a_0 = 0$.
Dla jakich wartości n istnieją ogólne wzory na rozwiązanie tego równania? Sprawdź w Mathematica® dla kilku
wybranych wartości n.

In[•]:=
```
IndexToLetter[index_] := FromCharacterCode[First[ToCharacterCode["a"]] + index]
```

In[•]:=
```
Polynomial[n_] := Total[Map[k ↦ IndexToLetter[k] * x^(n - k), Range[0, n]]]
```

In[•]:=
```
Polynomial[7]
```

Out[•]=
$$h + g\,x + f\,x^2 + e\,x^3 + d\,x^4 + c\,x^5 + b\,x^6 + a\,x^7$$

In[•]:=
```
Solve[Polynomial[2] == 0, x]
```

Out[•]=
$$\left\{\left\{x \to \frac{-b-\sqrt{b^2-4\,a\,c}}{2\,a}\right\},\ \left\{x \to \frac{-b+\sqrt{b^2-4\,a\,c}}{2\,a}\right\}\right\}$$

In[•]:=
```
Solve[Polynomial[4] == 0, x]
```

Out[•]=
$$\left\{\left\{x \to -\frac{b}{4\,a}-\frac{1}{2}\sqrt{\frac{b^2}{4\,a^2}-\frac{2\,c}{3\,a}+\frac{2^{1/3}\left(c^2-3\,b\,d+12\,a\,e\right)}{3\,a\left(\langle\!\langle 7\rangle\!\rangle+\sqrt{-4\left(\langle\!\langle 1\rangle\!\rangle\right)^3+\left(\langle\!\langle 1\rangle\!\rangle\right)^2}\right)^{1/3}}+\frac{\left(2\,c^3-9\,b\,c\,d+27\,a\,d^2+27\,b^2\,e-72\,a\,c\,e+\sqrt{-4\left(c^2-3\,b\,d+12\,a\,e\right)^3+\left(\langle\!\langle 1\rangle\!\rangle\right)^2}\right)^{1/3}}{3\times 2^{1/3}\,a}}-\right.$$
$$\frac{1}{2}\sqrt{\frac{b^2}{2\,a^2}-\frac{4\,c}{3\,a}-\frac{2^{1/3}\left(c^2-3\,b\,d+12\,a\,e\right)}{3\,a\left(2\,c^3-9\,b\,c\,d+27\,a\,d^2+27\,b^2\,e-72\,a\,c\,e+\sqrt{-4\left(c^2-3\,b\,d+12\,a\,e\right)^3+\left(\langle\!\langle 1\rangle\!\rangle\right)^2}\right)^{1/3}}-$$
$$\left.\frac{\left(2\,c^3-9\,b\,c\,d+27\,a\,d^2+27\,b^2\,e-72\,a\,c\,e+\sqrt{-4\left(c^2-3\,b\,d+12\,a\,e\right)^3+\left(2\,c^3-9\,b\,c\,d+\langle\!\langle 1\rangle\!\rangle+27\,b^2\,e-72\,a\,c\,e\right)^2}\right)^{1/3}}{3\times 2^{1/3}\,a}-\right.$$
$$\left.\frac{-\frac{b^3}{a^3}+\frac{4\,b\,c}{a^2}-\frac{8\,d}{a}}{4\sqrt{\frac{b^2}{4\,a^2}-\frac{2\,c}{3\,a}+\frac{2^{1/3}\left(c^2-3\,b\,d+12\,a\,e\right)}{3\,a\left(\langle\!\langle 7\rangle\!\rangle+\sqrt{-4\left(\langle\!\langle 1\rangle\!\rangle\right)^3+\left(\langle\!\langle 1\rangle\!\rangle\right)^2}\right)^{1/3}}+\frac{\left(2\,c^3-9\,b\,c\,d+27\,a\,d^2+27\,b^2\,e-72\,a\,c\,e+\sqrt{-4\left(c^2-3\,b\,d+12\,a\,e\right)^3+\left(\langle\!\langle 1\rangle\!\rangle\right)^2}\right)^{1/3}}{3\times 2^{1/3}\,a}}}\right\},$$
$$\left\{x \to \langle\!\langle 1\rangle\!\rangle\right\},\ \left\{x \to \langle\!\langle 1\rangle\!\rangle\right\},\ \left\{x \to -\frac{b}{4\,a}+\frac{1}{2}\sqrt{\langle\!\langle 4\rangle\!\rangle+\frac{\langle\!\langle 1\rangle\!\rangle\ \langle\!\langle 1\rangle\!\rangle}{3\times 2^{\langle\!\langle 1\rangle\!\rangle}\ a}}+\right.$$
$$\left.\frac{1}{2}\sqrt{\frac{b^2}{2\,a^2}-\frac{4\,c}{3\,a}-\frac{2^{1/3}\left(c^2-3\,b\,d+12\,a\,e\right)}{3\,a\left(\langle\!\langle 1\rangle\!\rangle\right)^{1/3}}-\frac{\left(\langle\!\langle 1\rangle\!\rangle\right)^{1/3}}{3\times 2^{1/3}\,a}+\frac{-\frac{b^3}{a^3}+\frac{4\,b\,c}{a^2}-\frac{8\,d}{a}}{4\sqrt{\frac{b^2}{4\,a^2}-\frac{2\,c}{3\,a}+\frac{2^{\langle\!\langle 1\rangle\!\rangle}\ \langle\!\langle 1\rangle\!\rangle}{3\,a\ \langle\!\langle 1\rangle\!\rangle}+\frac{\left(\langle\!\langle 1\rangle\!\rangle\right)^{1/3}}{3\times 2^{1/3}\,a}}}\right\}\right\}$$

Full expression not available (original memory size: 103.6 kB)

In[•]:=
```
Solve[Polynomial[5] == 0, x]
```

Out[•]=
$$\left\{\left\{x \to \text{Root}\left[f+e\,\#1+d\,\#1^2+c\,\#1^3+b\,\#1^4+a\,\#1^5\ \&,\ 1\right]\right\},\right.$$
$$\left\{x \to \text{Root}\left[f+e\,\#1+d\,\#1^2+c\,\#1^3+b\,\#1^4+a\,\#1^5\ \&,\ 2\right]\right\},$$
$$\left\{x \to \text{Root}\left[f+e\,\#1+d\,\#1^2+c\,\#1^3+b\,\#1^4+a\,\#1^5\ \&,\ 3\right]\right\},$$
$$\left\{x \to \text{Root}\left[f+e\,\#1+d\,\#1^2+c\,\#1^3+b\,\#1^4+a\,\#1^5\ \&,\ 4\right]\right\},$$
$$\left.\left\{x \to \text{Root}\left[f+e\,\#1+d\,\#1^2+c\,\#1^3+b\,\#1^4+a\,\#1^5\ \&,\ 5\right]\right\}\right\}$$

```
In[•]:= Solve[Polynomial[6] == 0, x]
```

```
Out[•]= {{x → Root[g + f #1 + e #1^2 + d #1^3 + c #1^4 + b #1^5 + a #1^6 &, 1]},
  {x → Root[g + f #1 + e #1^2 + d #1^3 + c #1^4 + b #1^5 + a #1^6 &, 2]},
  {x → Root[g + f #1 + e #1^2 + d #1^3 + c #1^4 + b #1^5 + a #1^6 &, 3]},
  {x → Root[g + f #1 + e #1^2 + d #1^3 + c #1^4 + b #1^5 + a #1^6 &, 4]},
  {x → Root[g + f #1 + e #1^2 + d #1^3 + c #1^4 + b #1^5 + a #1^6 &, 5]},
  {x → Root[g + f #1 + e #1^2 + d #1^3 + c #1^4 + b #1^5 + a #1^6 &, 6]}}
```

Ciekawostka: Na powyższe pytanie odpowiada Twierdzenie Abela-Ruffiniego

---

Zadanie 4. Rozwiązywanie równań, wielomiany wyższych stopni, `Product`, `Expand`, `Table`, oraz inne

Utworzyć wielomian 10-stopnia, $h(x)$, o miejscach zerowych $x1, x2, \ldots, xn \in \mathbb{R}$ wylosowanych z rozkładu jednorodnego na przedziale [–1, 1].
W tym celu wylosuj miejsca zerowe korzystając z funkcji `RandomReal`, a następnie skorzystaj z funkcji `Product` by utworzyć wielomian $h(x)$ o zadanych miejscach zerowych.

```
In[•]:= ? RandomReal
```

```
Out[•]=
Symbol

RandomReal[] gives a pseudorandom real number in the range 0 to 1.

RandomReal[{x_min, x_max}] gives a pseudorandom real number in the range x_min to x_max.

RandomReal[x_max] gives a pseudorandom real number in the range 0 to x_max.

RandomReal[range, n] gives a list of n pseudorandom reals.

RandomReal[range, {n1, n2, ...}] gives an n1×n2×... array of pseudorandom reals.
```

```
In[•]:= roots = RandomReal[{-1, 1}, 10]
```

```
Out[•]= {0.0678822, 0.232231, -0.512611, 0.768357, -0.492279,
  -0.877876, 0.281027, -0.446885, -0.532166, -0.0804542}
```

```
In[•]:= h[x_] := Product[x - roots[[i]], {i, Length[roots]}]
```

Następnie należy:

1) Wypisać wielomian $h(x)$ korzystając z polecenia `Expand`[$h(x)$]

```
In[•]:= Expand[h[x]]
```

```
Out[•]= 0.0000144282 - 0.0000321714 x - 0.00300971 x^2 + 0.0001818 x^3 + 0.0705109 x^4 +
  0.0859219 x^5 - 0.473687 x^6 - 1.08451 x^7 + 0.0219418 x^8 + 1.59277 x^9 + x^10
```

2) Sprawdzić ile rozwiązań będzie miało równanie $h(x) = a$

gdzie a jest parametrem zmieniającym się w wąskim zakresie od –0.01 do 0.01, co 0.0005.

```
In[•]:= aValues = Range[-0.01, 0.01, 0.0005]
```

```
Out[•]= {-0.01, -0.0095, -0.009, -0.0085, -0.008, -0.0075, -0.007, -0.0065, -0.006, -0.0055,
 -0.005, -0.0045, -0.004, -0.0035, -0.003, -0.0025, -0.002, -0.0015, -0.001,
 -0.0005, 0., 0.0005, 0.001, 0.0015, 0.002, 0.0025, 0.003, 0.0035, 0.004, 0.0045,
 0.005, 0.0055, 0.006, 0.0065, 0.007, 0.0075, 0.008, 0.0085, 0.009, 0.0095, 0.01}
```

Wyniki zachować jako lista `nSol`. Można użyć funkcji `Table` do zapełniania listy `nSol`.

```
In[•]:= nSol = Map[
  a ↦ {a, Length[Solve[h[x] == a, {x}, Reals]]},
  aValues
 ]
```

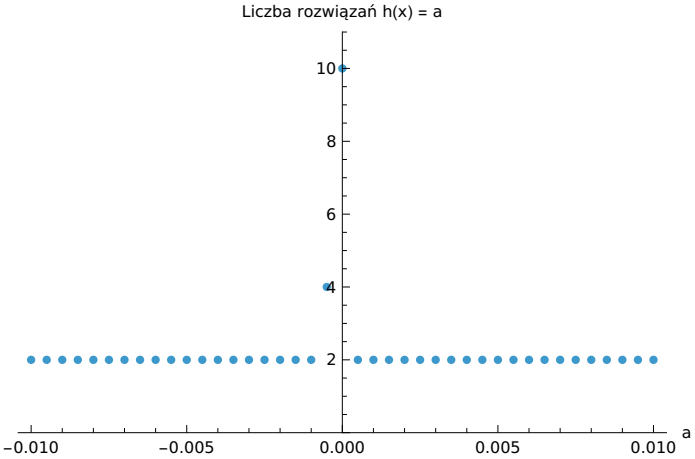
```
Out[•]= {{-0.01, 2}, {-0.0095, 2}, {-0.009, 2}, {-0.0085, 2}, {-0.008, 2}, {-0.0075, 2}, {-0.007, 2},
 {-0.0065, 2}, {-0.006, 2}, {-0.0055, 2}, {-0.005, 2}, {-0.0045, 2}, {-0.004, 2}, {-0.0035, 2},
 {-0.003, 2}, {-0.0025, 2}, {-0.002, 2}, {-0.0015, 2}, {-0.001, 2}, {-0.0005, 4},
 {0., 10}, {0.0005, 2}, {0.001, 2}, {0.0015, 2}, {0.002, 2}, {0.0025, 2}, {0.003, 2},
 {0.0035, 2}, {0.004, 2}, {0.0045, 2}, {0.005, 2}, {0.0055, 2}, {0.006, 2}, {0.0065, 2},
 {0.007, 2}, {0.0075, 2}, {0.008, 2}, {0.0085, 2}, {0.009, 2}, {0.0095, 2}, {0.01, 2}}
```

3) Przedstawić wartości `nSol` na wykresie korzystając z `ListPlot`

Upewnić się, że zakres wykresu obejmuje wszystkie punkty.
Zadbać o prezentację wyniku, zapoznać się z opcjami `ListPlot` przez **F1** albo `Options[ListPlot]`.

In[•]:=
```
ListPlot[
 nSol,
 PlotRange → {0, Length[roots] + 1},
 AxesLabel → {"a", "Liczba rozwiązań h(x) = a"}
]
```

Out[•]=



4) Zautomatyzować proces generowania wykresu

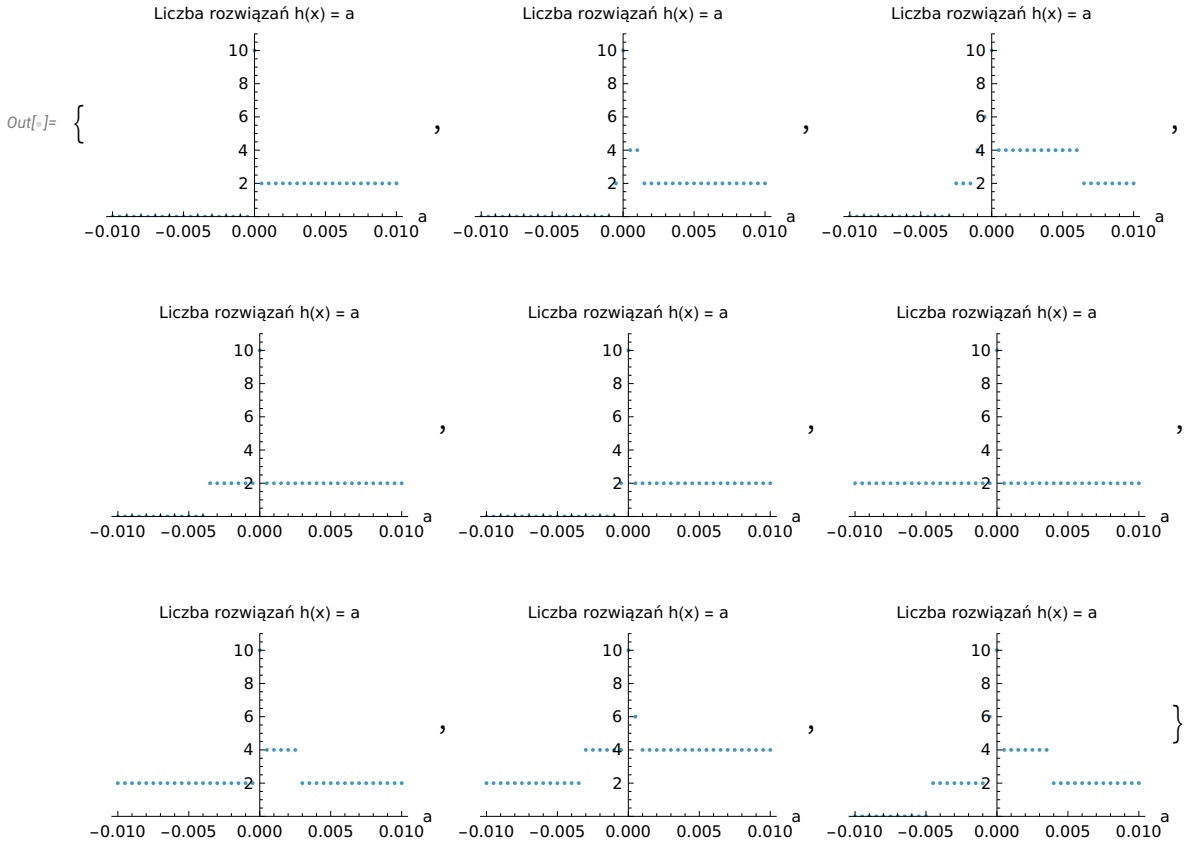
Zautomatyzować proces generowania wykresu z punktu 3) tak by można było go otrzymać dla wielomianu $h(x)$ z nowo wylosowanymi miejscami zerowymi.

In[•]:=
```
GeneratePolynomialAndPlot[] := Module[
  {},
  roots = RandomReal[{-1, 1}, 10];
  h[x_] := Product[x - roots[[i]], {i, Length[roots]}];
  aValues = Range[-0.01, 0.01, 0.0005];
  nSol = Map[
    a ↦ {a, Length[Solve[h[x] == a, {x}, Reals]]},
    aValues
   ];
  ListPlot[
   nSol,
   PlotRange → {0, Length[roots] + 1},
   AxesLabel → {"a", "Liczba rozwiązań h(x) = a"}
  ]
 ];
```

Następnie utworzyć listę `pSol` zawierającą wykresy jak w punkcie 3) dla 9-ciu różnych wielomianów $h(x)$ z wylosowanymi miejscami zerowymi.

In[•]:=
```
pSol = Map[plot ↦ GeneratePolynomialAndPlot[], Range[9]]
```

Out[•]=



5) Zadbać o prezentację wyniku z punktu 4)

Zapoznać się z funkcją `Partition`, `TableForm`, `GraphicsGrid`.

In[•]:= `? Partition`

Out[•]=

Symbol

Partition[*list*, *n*] partitions *list* into nonoverlapping sublists of length *n*.

Partition[*list*, *n*, *d*] generates sublists with offset *d*.

Partition[*list*, {n_1, n_2, ...}] partitions a nested list into blocks of size $n_1 \times n_2 \times \ldots$.

Partition[*list*, {n_1, n_2, ...}, {d_1, d_2, ...}] uses offset d_i at level *i* in *list*.

Partition[*list*, *n*, *d*, {k_L, k_R}] specifies that the first element of *list* should appear at position k_L in the first sublist, and the last element of *list* should appear at or after position k_R in the last sublist. If additional elements are needed, Partition fills them in by treating *list* as cyclic.

Partition[*list*, *n*, *d*, {k_L, k_R}, *x*] pads if necessary by repeating the element *x*.

Partition[*list*, *n*, *d*, {k_L, k_R}, {x_1, x_2, ...}] pads if necessary by cyclically repeating the elements x_i.

Partition[*list*, *n*, *d*, {k_L, k_R}, {}] uses no padding, and so can yield sublists of different lengths.

Partition[*list*, *nlist*, *dlist*, {$klist_L$, $klist_R$}, *padlist*] specifies alignments and padding in a nested list.

In[•]:= `? TableForm`

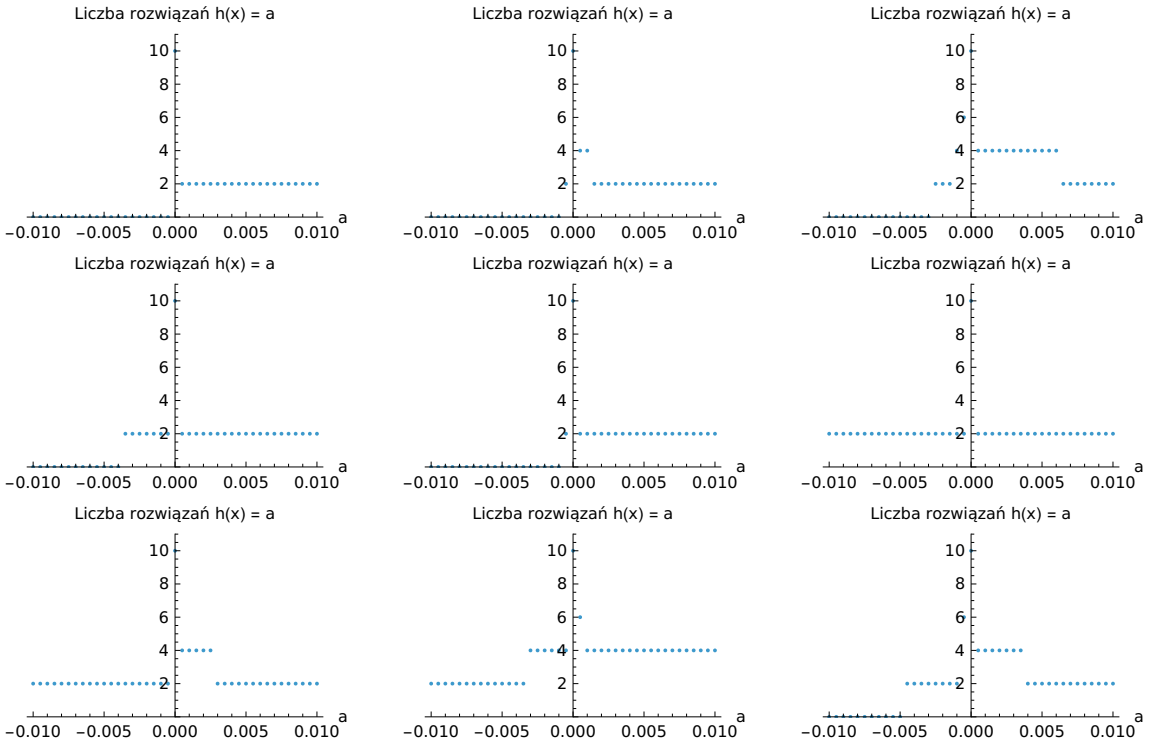
Out[•]=

Symbol

TableForm[*list*] prints with the elements of *list* arranged in an array of rectangular cells.

In[•]:=
```
partitioned = Partition[pSol, 3];
TableForm[partitioned]
```

Out[•]//TableForm=



Zadanie 5. Równania transcendentne, Solve, NSolve, FindRoot

Znaleźć rozwiązania równania: *3 sin(7x) - 2 ln(x) = 0*

In[•]:=
```
d[x_] := 3 Sin[7 x] - 2 Log[x]
```

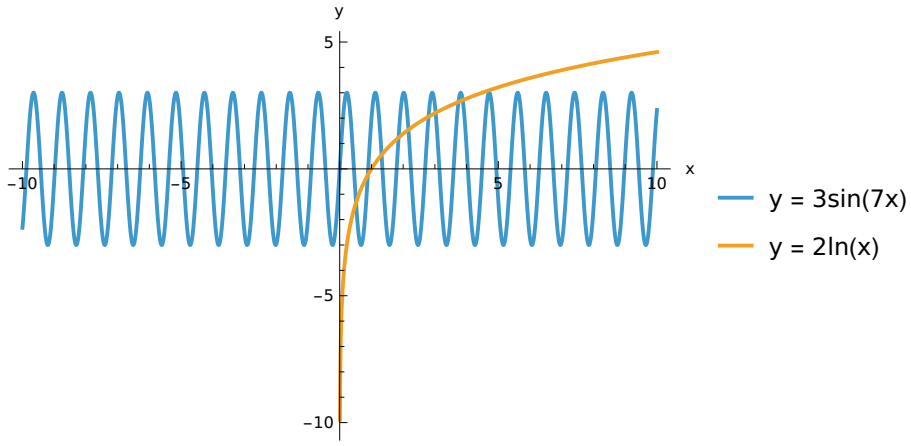
Korzystając z:

1) funkcji Solve z wybranym zakresem x

Wskazówka: By wybrać odpowiedni zakres narysuj na wykresie funkcje 3 sin(7x) oraz 2 ln x.

In[•]:=
```
Plot[
 {3 Sin[7 x], 2 Log[x]},
 {x, -10, 10},
 PlotLegends → {"y = 3sin(7x)", "y = 2ln(x)"},
 AxesLabel → {"x", "y"}
]
```

Out[•]=



In[•]:=
```
Solve[{d[x] == 0, 1/4 ≤ x ≤ 5}, Reals]
```

Out[•]=
```
{{x → 0.514...}, {x → 0.886...}, {x → 1.32...}, {x → 1.86...},
 {x → 2.17...}, {x → 2.80...}, {x → 3.02...}, {x → 3.74...}, {x → 3.88...}}
```

Co zwraca Solve, bez wybranego zakresu?

Jaki komunikat zwraca Mathematica®?

In[•]:=
```
Solve[d[x] == 0, Reals]
```

Solve: This system cannot be solved with the methods available to Solve. Try Reduce or FindInstance instead.

Out[•]=
```
Solve[-2 Log[x] + 3 Sin[7 x] == 0, ℝ]
```

2) NSolve

In[•]:=
```
? NSolve
```

Out[•]=

Symbol

NSolve[*expr*, *vars*] attempts to find numerical approximations to the solutions of the system *expr* of equations or inequalities for the variables *vars*.

NSolve[*expr*, *vars*, Reals] finds solutions over the domain of real numbers.

In[•]:=
```
nSolveResult = NSolve[d[x] == 0, Reals]
```

Out[•]=
```
{{x → 0.514391}, {x → 0.886065}, {x → 1.31982}, {x → 1.8559},
 {x → 2.16663}, {x → 2.80091}, {x → 3.02311}, {x → 3.74412}, {x → 3.87805}}
```

Ile rozwiązań zwraca ta funkcja?

In[•]:=
```
Length[nSolveResult]
```

Out[•]= 9

3) FindRoot

In[•]:=
```
? FindRoot
```

Out[•]=

Symbol

FindRoot[f, {x, x_0}] searches for a numerical root of f, starting from the point $x = x_0$.

FindRoot[*lhs* == *rhs*, {x, x_0}] searches for a numerical solution to the equation *lhs* == *rhs*.

FindRoot[{f_1, f_2, ...}, {{x, x_0}, {y, y_0}, ...}] searches for a simultaneous numerical root of all the f_i.

FindRoot[{eqn_1, eqn_2, ...}, {{x, x_0}, {y, y_0}, ...}]

searches for a numerical solution to the simultaneous equations eqn_i.

In[•]:=
```
findRootResult = FindRoot[d[x], {x, 1}]
```

Out[•]=
```
{x → 0.886065}
```

Ile rozwiązań zwraca ta funkcja?

In[•]:=
```
Length[findRootResult]
```

Out[•]= 1